MUHAMMAD Qutab

AI / Machine Learning Engineer

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https://github.com/MuhammadQutab https://muhammadqutab.github.io/

Profile Summary

Results-driven AI/ML professional with hands-on experience in computer vision, NLP, and predictive analytics. Gold Medalist in BS Artificial Intelligence, skilled in developing, training, and deploying machine learning models. Proven track record in executing impactful AI projects, including robotic automation, deep learning applications, and data-driven forecasting. Adept at problem-solving and delivering scalable AI solutions.

Education

University of Kotli Azad Kashmir **Dec 2020 – Feb 2025**

Bachelor of Science in Artificial Intelligence (BS AI) *Kotli, AJK*

Coursework

- Computer Vision
- Deep Learning
- Machine Learning
- Artificial Intelligence
- Algorithms Analysis
- Database Management
- Python Programming
- Power BI

Experience

Tech Fellow **IMSB (I-VIII) Satra Meel (FA)**

AI/Robotics Trainer *Oct 2025 – Feb 2026*

- Delivered practical training in Artificial Intelligence, Robotics, and Algorithms to school students through interactive STEAM-based learning activities.
- Conducted hands-on sessions using Arduino, sensors, and basic automation projects to develop problem-solving and computational thinking skills.
- Guided students in understanding AI fundamentals, logical reasoning, and real-world robotics applications in an engaging classroom environment.

Online Tutor | AS Levels / O Levels **Eahana Online Academy**

Computer Science Tutor *Oct 2025 – Feb 2026*

- Taught Computer Science concepts to AS Level and O Level students through structured online sessions and personalised academic support.
- Simplified programming, algorithms, and theoretical computing topics to improve student understanding and examination performance.
- Prepared practice materials, guided assignments, and monitored student progress through regular feedback and mentoring.

Projects

RoboTechArm (FYP) | Python, YOLOv8, OpenCV, Flask, Firebase, Arduino

- Final Year Project – developed an intelligent robotic arm capable of detecting and picking fruits using YOLOv8 object detection.
- Integrated Firebase for real-time command communication and built a Flask web app for control interface.
- Enabled automatic motion response only when a commanded fruit was detected, ensuring precision and smart interaction.
- *GitHub:* https://github.com/MuhammadQutab/RoboTechArm

Churn Intelligence App (MSc Research Support) | Python, XGBoost, SHAP, Streamlit

- Supported MSc pipeline: Online Retail II data, customer features, XGBoost churn prediction and SHAP insights.
- Helped turn risk/value segments into retention priorities and financial scenarios (holdout ROC-AUC: 0.819).
  GitHub: https://github.com/MuhammadQutab/Churn-Intelligence-App

Student Academic Risk Prediction (MSc Research Support) | Scikit-learn, SHAP, Streamlit

- Guided early-warning MSc pipeline: student features (excluding G1/G2), tuned Logistic Regression and a 0.45 threshold.
- Supported SHAP explanations, individual risk scores and an interactive dashboard with student support actions.
  GitHub: https://github.com/MuhammadQutab/student-risk-streamlit-app

Geo-Aware Intrusion Detection (MSc Research Support) | Python, XGBoost, Streamlit

- Mentored MSc model comparison on CSE-CIC-IDS2018; XGBoost classified benign traffic and four attack types.
- Supported CSV/bulk inference, confidence/severity analysis and contextual geo-aware alerts in a SOC dashboard.
  GitHub: https://github.com/MuhammadQutab/ids-threat-detection-dashboard

CityMate AI (Independent Project) | Next.js, TypeScript, Gemini API, OpenStreetMap

- Led a bilingual AI city assistant for everyday needs, local services and newcomers across pilot Pakistani cities.
- Designed a grounded Plan > Execute > Narrate agent with real map results, service fallbacks and Urdu support.
- Got 1st position and token of appreciation in a virtual hackathon organized by Global Engineering Network for Innovation and Intelligence.
  GitHub: https://github.com/MuhammadQutab/citymate-ai

Certifications

AI for everyone — Coursera - Dec 2023
STEAM Trainer — LearnOBots (NSTP) - Oct 2025
Certificate of Innovation and Excellence — GEN_II – Oct 2026

Technical Skills

Programming & Languages: Python, SQL, Power BI
Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, XGBoost, SHAP, OpenCV, Pandas, NumPy, YOLOv8, Flask, Streamlit
Machine Learning & AI: Deep Learning, Computer Vision, Natural Language Processing (NLP), Transfer Learning, Model Training & Evaluation, Data Preprocessing, Feature Engineering
Tools & Platforms: Jupyter Notebook, VS Code, Google Colab, GitHub, Power BI, Arduino IDE, MySQL, Linux (Kali), Windows
Specializations: Computer Vision, Machine Learning, Deep Learning, AI-Powered Robotics, Explainable AI

Leadership / Extracurricular

SEC & NESCON (Kotli AJK, FAST University) — **Dec 2023 - Dec 2024**
Team Lead/ Ambassador — *University of Kotli AJK & FAST University Islamabad*

- Became a part of 2 high-impact events, including the Startup and Entrepreneurial Conference at University of Kotli AJK and NESCON at FAST University Islamabad.
- Led a team of 5+, driving innovations and ensuring successful execution.
- Engaged talented people to do the right job.